PHYSICAL CUME
OCTOBER 19, 2001

For the following processes determine whether each of the thermodynamic quantities listed is greater than, equal to, or less than zero for the *system* described. Consider all gases to behave ideally. The system is shown in italic type in each case. Indicate your reasoning and state explicitly any reasonable assumptions or approximations that you need to make.

(a) A *sample of gas* is carried through a complete Carnot cycle (isothermal expansion, adiabatic expansion, isothermal compression, adiabatic compression—all reversible): ΔT, q, w, ΔE, ΔH, ΔS, ΔG.

(b) A sample of *hot water* is mixed with a sample of *cold water* in a thermally insulated, closed container of fixed volume: w, q, ΔE, ΔH, ΔS.

(c) An *ideal gas* expands adiabatically and reversibly: ΔV, ΔT, w, q, ΔE, ΔH, ΔS.

(d) A flask of *liquid nutrient solution inoculated with a small sample of bacteria* is maintained for several days in a thermostat until the bacteria have multiplied 1000-fold: ΔT, w, q, ΔE, ΔH, ΔG.